Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/17/2020
PI Name:	Oates, Chris Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2021 and new project established due to change in Principal Investigator, per PI and U. Israelsson/JPL (Ed., 9/13/21) NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2022 per U. Israelsson/JPL (Ed., 9/13/18) NOTE: End date changed to 11/30/2018 per M. Lee/NASA HQ (Ed., 2/16/16)		
Key Personnel Changes/Previous PI:	New Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; see new project, "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Patla)" for continuation. September 2020 report: Dr. Steven Jefferts is no longer a CoInvestigator on this project. ED. NOTE (December 2015): PI Chris Oates took over the project from Thomas O'Brian on 11/16/2015.		
COI Name (Institution):			
Grant/Contract No.:	NNH12AT81I		
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Task Description:

NOTE: NEW Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; see new project, "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Patla)"
EDITOR'S NOTE (December 2015): Principal Investigator (PI) Chris Oates took over the project from Thomas O'Brian on 11/16/2015.

To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10^-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better.

• Test of gravitation redshift at 2 x 10^-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19.

• Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18.

• Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16.

• A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time.

Gap 12 Mo. (2018-9): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2019-2020): Assuming successful launch in 2021, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Evaluate and improve the performance of the new frequency-based time scale optimized for ACES frequency comparisons. $80k for partial support of staff time.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.
Task Progress:	In the most recent reporting period (October 2019 through September 2020) work has focused on three main tasks: 1. Development of a new frequenct-based NIST Time Scale that will connect directly to the ACES Microwave Link Ground Terminal (MWL GT) 2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics 3. Consolidating the data analysis infrastructure and corrections required for the ACES frequency data Progress on each task is described below. 1. Development of a new frequency-based NIST Time Scale that will connect directly to the ACES Microwave Link Ground Terminal (MWL GT) The basis for the ACES program will be the advanced primary frequency standard on the International Space Station, which will link to participating microwave ground stations around the world. NIST has yet to receive its Microwave Link Ground Terminal (MWL GT) from ESA and is awaiting confirmation of an expected delivery date. NIST has been closely coordinating with ESA on details of the MWL GT hardware, information technology, and infrastructure requirements. In the meantime NIST clock scientists have finished putting in place the required infrastructure for connecting the ACES clock signals to NIST's most stable microwave sources. Over the past year, the main focus has been to upgrade the microwave time sources themselves to ones better suited for ACES clock frequency comparisons. In particular, the current version of the NIST timescale is designed to be "on time" with respect to international atomic time (TAI). This implies that the timescale algorithm is allowed to take frequency steps and jumps with the requirement that the time steps are allowed are small. However, for ACES, it would be better to have a time scale based on a stable frequency source. Indeed, having a continuous frequency output based on the ensemble of clocks with almost 100% duty cycle would enable us to compare two non-collocated clocks very accurately. This would in turn enable us to test (and constrain) frequency transfer noise and more readily participate in frequency-based tests of fundamental physics. To this end, we initiated a project to upgrade the time scale itself by, (i) constructing a new, frequency-based time scale (details below), and (ii) locating this new time scale nearer the Ground Terminal to minimize timing noise in the links. The main effort over the past year has been to continue this upgrade with a focus on beta-testing the initial design along with its algorithms. In particular we analyzed time scale data over a 28-month period for three different masers steered by the new time scale algorithms. These results indicate that with current steering algorithms we are able to produce signals that are free of significant time or frequency jumps while maintaining timing offsets relative to UTC that are within acceptable limits (e.g., ~ few ns). Such signals will be ideal for ACES comparisons in that frequency comparisons will not be contaminated by frequency steps during the averaging times. We plan to continue to optimize the algorithms to improve performance. 2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics NIST continues to improve the performance of its multiple optical and microwave frequency standards, which will be crucial to international time and frequency comparisons and ACES-mediated fundamental research. The past project year included advances in several of NIST's frequency standards: • NIST performed frequency comparisons between its Yb and Sr lattice clocks and its Al+ logic clocks through optical fiber and free space links. These optical atomic clocks all have evaluated fractional frequency uncertainty budgets below 2 x 10^-18, the lowest values to date worldwide. These comparisons yielded optical frequency ratios with uncertainties of ~7 x 10^-18, the lowest to date by nearly an order of magnitude. These measurements represent an important step towards the redefinition of the SI (International System of Units) second in terms of an optical transition. Moreover, the high level of stability demonstrated by these clocks dramatically advances the goals of ACES time and frequency transfer given the relatively short visibility of the International Space Station (ISS) to any MWL GT. • The Al+ logic clock and the Sr lattice clocks at NIST reported new absolutely frequency measurements with an

uncertainty below 3.6 x 10^-16, virtually limited by the microwave clocks that are currently used to realize the SI second. Moreover, NIST reported Yb frequencies to the BIPM (Bureau International des Poids et Mesures) in its first efforts towards use of its optical frequency standards as contributors to UTC.

At the time of this report, NIST (including NIST/JILA) currently operates world class microwave primary frequency standards (Cs fountains) at the 10^-16 level and three leading optical standards (Yb and Sr neutral atom lattice clocks, Al ion clocks) at the 10^-18 level. Other laboratories across the world are achieving high performance as well, and the NIST optical frequency standards are likely to continue to improve. This dramatic improvement of optical frequency standard performance strongly supports ACES goals of improved international time and frequency transfer and the ability to improve fundamental tests of physics.

Additionally, we note that NIST has initiated the construction of a next generation time scale that will be based on these advanced optical (rather than microwave) sources. While not designed specifically for the ACES mission, this future time scale could be online for part of the ACES mission and could offer performance more than an order of magnitude beyond existing time scales due to the superior stability of optical standards.

3. Consolidating the data analysis infrastructure and corrections required for the ACES frequency data

An important part of this overall effort has been to be certain that all necessary corrections to the ACES signals have been accounted for. We recently consolidated all the theoretical material generated so far in a single location. This file contains a detailed derivation of the spatial coordinate/time transformations from the Earth-Centered Inertial Coordinate (ECI) frame to the freely falling frame of the ISS or of some other orbiting satellite. The calculation is taken to third order in the quantity 1/c, where c is the speed of light. To this order, the derivation is valid even for an optical clock in an Earth-bound satellite because fourth order terms would be only of order a few parts in 10^20. Summarizing the content of the file “transformation_to_local_frame.pdf,” the following quantities are derived:

1. Spatial and time coordinate transformations from the ECI to a freely falling (locally inertial) frame corresponding to the freely-falling center of mass of an earth-bound satellite; 2. Frequency in the ECI of a signal from a freely falling transmitter; 3. Measured proper frequency of a signal from a freely falling transmitter at a detector at rest in the ECI frame; 4. Confirmation that in the freely falling frame there are no linear terms in local coordinates that contribute to the metric tensor, (required by the principle of equivalence).

The next steps in this derivation involve the deflection of electromagnetic signals from a freely falling transmitter (finished but not reviewed yet) and computation of additional transformations due to (a) rotation of the spacecraft (for example, rotation once per orbit due to flying with the nose of the ISS pointed towards the nadir) and (b) transformations from a detector at rest in the ECI to one at the same place but fixed in the Earth-Centered Earth Fixed (ECEF) frame.

NOTE: NEW Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla. Project continues as "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Patla)"; see that project for future reporting.

Bibliography Type:	Description: (Last Updated: 04/05/2023)
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